A problem with image convolution is that a large number of calculations are involved. For instance, when a 512 by 512 pixel image is convolved with a 64 by 64 pixel PSF, more than a *billion* multiplications and additions are needed (i.e., $64 \times 64 \times 512 \times 512$). The long execution times can make the techniques impractical. Three approaches are used to speed things up.

The first strategy is to use a very small PSF, often only 3×3 pixels. This is carried out by looping through each sample in the output image, using optimized code to multiply and accumulate the corresponding nine pixels from the input image. A surprising amount of processing can be achieved with a mere 3×3 PSF, because it is large enough to affect the *edges* in an image.

The second strategy is used when a large PSF is needed, but its shape isn't critical. This calls for a filter kernel that is *separable*, a property that allows the image convolution to be carried out as a series of one-dimensional operations. This can improve the execution speed by *hundreds* of times.

The third strategy is FFT convolution, used when the filter kernel is large and has a specific shape. Even with the speed improvements provided by the highly efficient FFT, the execution time will be hideous. Let's take a closer look at the details of these three strategies, and examples of how they are used in image processing.

3×3 Edge Modification

Figure 24-4 shows several 3×3 operations. Figure (a) is an image acquired by an airport x-ray baggage scanner. When this image is convolved with a 3×3 delta function (a *one* surrounded by 8 zeros), the image remains unchanged. While this is not interesting by itself, it forms the baseline for the other filter kernels.

Figure (b) shows the image convolved with a 3×3 kernel consisting of a one, a negative one, and 7 zeros. This is called the **shift and subtract** operation, because a *shifted* version of the image (corresponding to the -1) is *subtracted* from the original image (corresponding to the 1). This processing produces the optical illusion that some objects are closer or farther away than the background, making a 3D or embossed effect. The brain interprets images as if the lighting is from *above*, the normal way the world presents itself. If the edges of an object are bright on the top and dark on the bottom, the object is perceived to be poking out from the background. To see another interesting effect, turn the picture upside down, and the objects will be pushed *into* the background.

Figure (c) shows an **edge detection** PSF, and the resulting image. Every edge in the original image is transformed into narrow dark and light bands that run parallel to the original edge. Thresholding this image can isolate either the dark or light band, providing a simple algorithm for detecting the edges in an image.